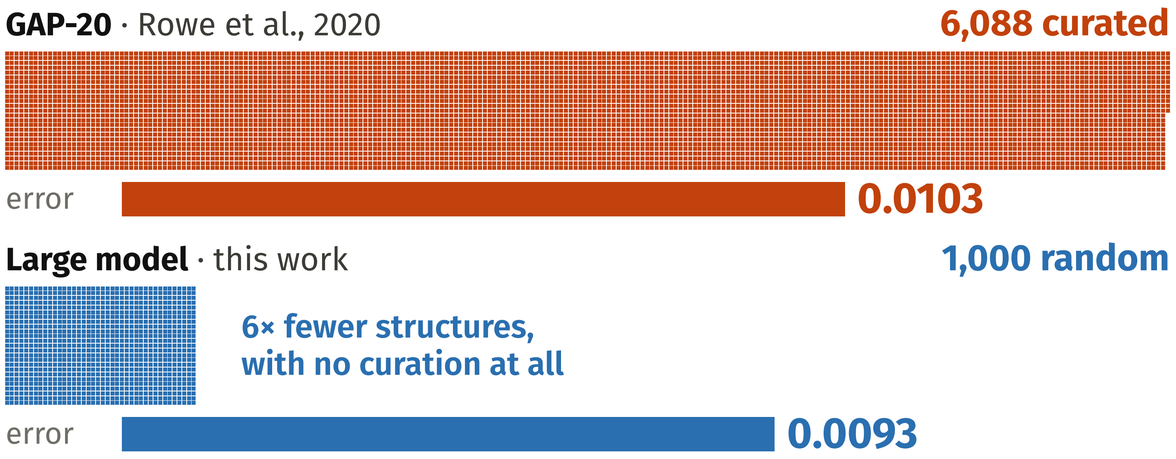
# A GAP potential for carbon from randomly selected, uncurated structures: comparison with GAP-20

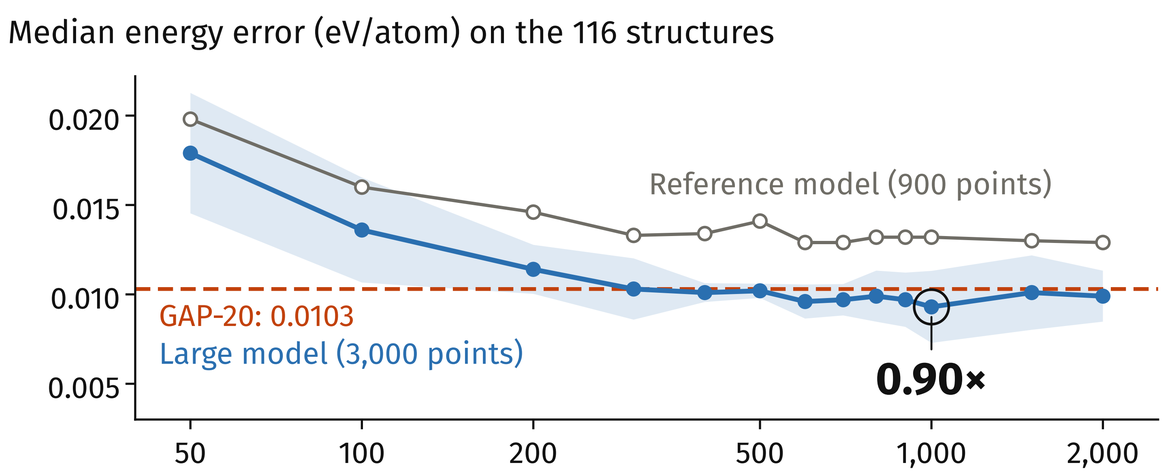
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Topic 5 · Computational Chemistry and Artificial Intelligence
Poster code: .....

**What accuracy does a machine-learning potential reach when its training structures are drawn at random?**

## With 1,000 random, uncurated structures, a GAP potential reaches the accuracy of GAP-20, which was trained on 6,088 curated ones.




Each square: one training structure. Error: median (eV/atom) on the **116 structures that neither model had seen**; ratio 0.90.

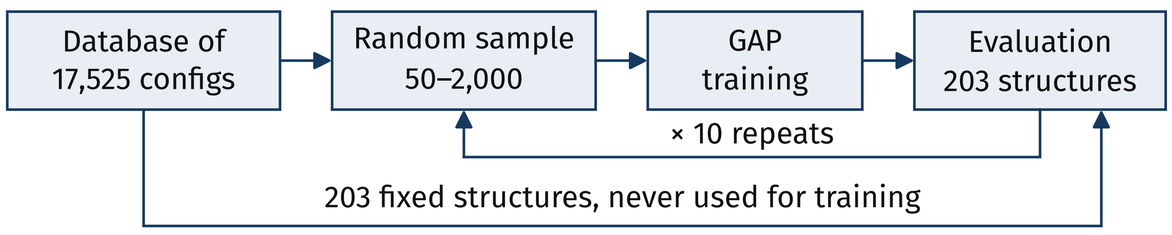



More accurate than GAP-20 at 9 of the 13 sizes; shading: ±1 standard deviation.

## 1 1,040 potentials on random samples of 50–2,000 structures

Machine-learning potentials (here GAP with the SOAP descriptor) give DFT energies at a cost orders of magnitude lower; their accuracy is set by the training structures.

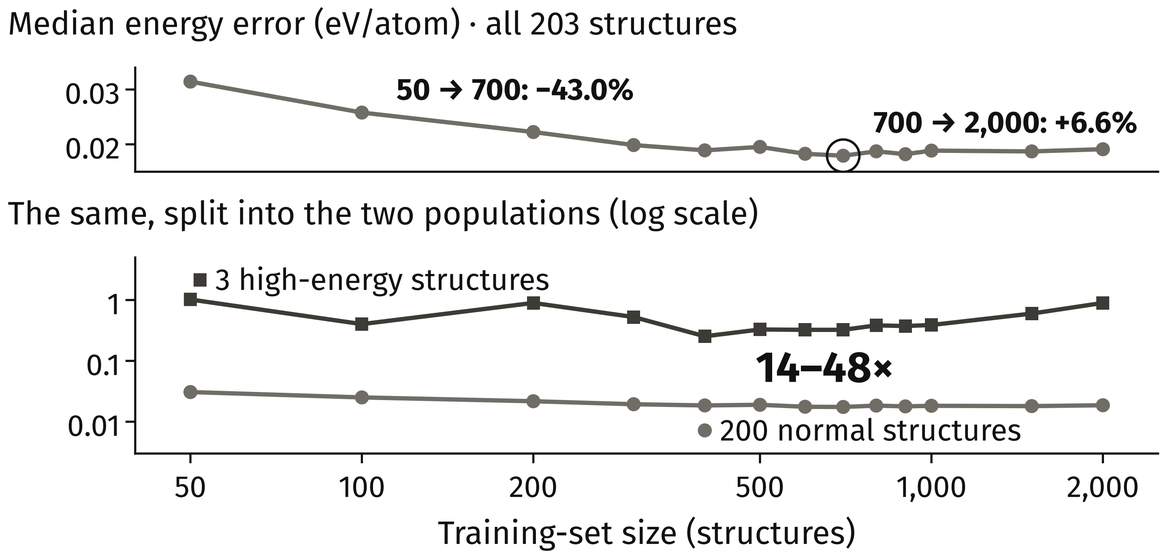
**8 settings × 13 sizes × 10 repeats** = 1,040



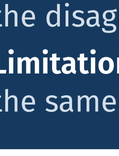

Each setting changes one parameter, e.g. the "capacity": 300–3,000 representative points. ~316 h on AUTh's "Aristotle" cluster (≈ 10,100 CPU hours).

Scan for the poster in Greek and English, the abstract and the references.
**9psxmpf.chalkidis.eu**

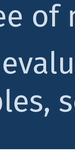
## 2 The curve hides 3 high-energy structures, 14–48× worse



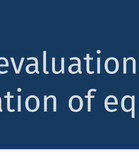

The rise is not overfitting (the training–validation gap closes, 7.6× → 1.1×): it comes from a few extreme structures, which are not predicted and which spoil the fit when they enter training.



**Crystalline**
10 atoms
−0.96 eV/atom



**Dimer**
2 atoms · 2.83 Å
+0.06 eV/atom

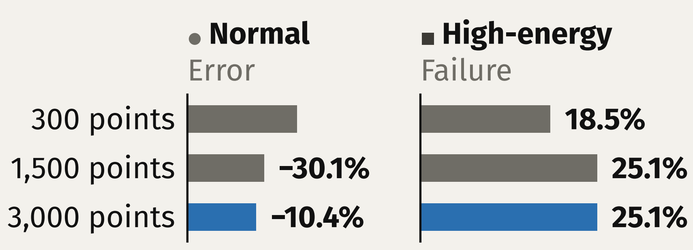


**Liquid**
216 atoms
+4.05 eV/atom

Criterion: DFT energy above −4 eV/atom (normal structures: median −7.20).

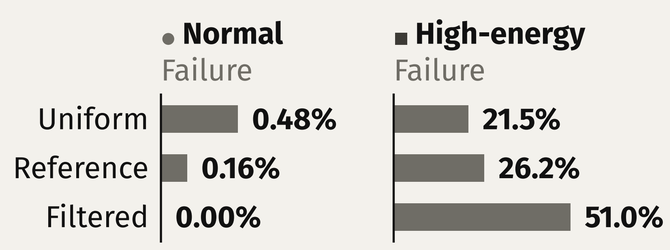
## 3 Three independent lines of evidence: the limit is in the data

### a A larger model: better on normal structures, not on high-energy ones



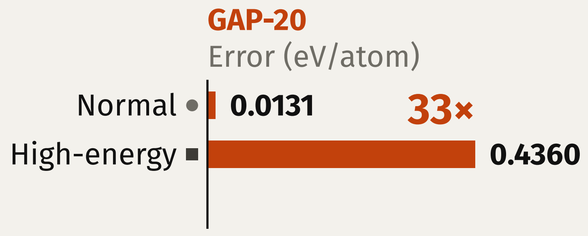

Overall −40.5% for 10× the points. Its error was predicted before training (0.0146–0.0166 eV/atom); the measured value was 0.0152.

### b Different training: what normal structures gain, high-energy ones lose




Failure: error > 1 eV/atom. Uniform: one tolerance for all structures; Filtered: no structures with a force > 50 eV/Å.

### c GAP-20 fails on the same structures




The same order of magnitude as our 14–48×, despite its curation and its 9,000 points.

**On an uncurated database, accuracy is limited by the data, not by the model.**
Curation is not a prerequisite, but one of the levers.

**Next step:** more high-energy structures in training, through active learning driven by the disagreement of a committee of models.

**Limitations:** only 3 high-energy evaluation structures; the evaluation set comes from the same database as our samples, so the 0.90 is an indication of equivalence.

Bartók, Payne, Kondor & Csányi (2010) *Phys. Rev. Lett.* 104, 136403 · Bartók, Kondor & Csányi (2013) *Phys. Rev. B* 87, 184115 · Rowe et al. (2020) *J. Chem. Phys.* 153, 034702
We thank the IT Center of AUTh for the "Aristotle" computing infrastructure.